Samuel Shilling

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RESEARCH INTERESTS

- Using Bayesian inference to model gamma-ray burst afterglows and associated astrophysical transients
- Using gamma-ray burst afterglows as cosmological probes of the early universe
- Understanding the physics driving empirical correlations observed in GRB afterglows
- Identifying and modeling off-axis afterglows detected with LSST

EMPLOYMENT

Researcher, Astrophysics **Jun. 2024 - Dec. 2024**
NASA Goddard Space Flight Center, MD, USA
Supervisor: Dr. Judith Racusin

EDUCATION

Ph.D., Astrophysics **Oct. 2023 - Apr. 2027**
Lancaster University, UK
Supervisor: Dr. Samantha Oates

MPhys (1st class), Physics and Astrophysics **Sep. 2019 - Jun. 2023**
University of Sheffield, UK
Supervisor: Prof. Clive Tadhunter

RELEVANT TECHNICAL SKILLS

- Scientific computing using Python, Linux and Git
- Developing data processing pipelines
- Quantitative data analysis and data visualisation
- Statistical inference and uncertainty estimation
- Simulating astrophysical transient light curves
- Bayesian inference for numerical model fitting

ADDITIONAL ACTIVITIES

- Postgraduate teaching assistant for undergraduate physics courses at Lancaster University (3 years)
- Chaired the Lancaster University astrophysics journal club in rotation (3 years, ongoing)
- Co-supervised masters student projects at Lancaster University (1 year, ongoing)
- Swift/UVOT burst scientist (3 years)
- Burst advocate scientist for stargate (2 months, ongoing)
- Author or co-author of 36 GCN Circulars for GRB followup observations

REFERENCES

Dr. Samantha Oates
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Dr. Judith Racusin
Astrophysics Science Division
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Dr. Gavin Lamb
Astrophysics Research Institute
Liverpool John Moores University
Liverpool
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CONFERENCES

- *Measurements of the geometric properties of GRBs and their implications on the luminosity—decay correlation.* **Oral presentation** at the UK transients meeting, 2026. University of Bath, UK.
- *Measurements of the geometric properties of GRBs and their implications on the luminosity—decay correlation.* **Oral presentation** at the Cosmic Explosions meeting, 2026. University of Cambridge, UK.
- *The intrinsic luminosity–decay correlation in subsamples of GRB X-ray afterglows.* **Oral presentation** at the National Astronomy Meeting, 2026. University of Birmingham, UK.
- *Evidence of an intrinsic luminosity-decay correlation in GRB radio light curves.* **Poster presentation** at the 11th Fermi symposium, 2024. University of Maryland, College Park, MD, USA.
- *An intrinsic luminosity-decay correlation in GRB radio light curves.* **Oral presentation** at the UK Transients meeting, 2024. University of Leicester, UK.

PUBLICATIONS

First author

- Shilling et al. (in prep.) *Inferring GRB jet structures, physical parameters, and geometric properties from a large sample of X-ray afterglows.* In preparation (est. 2026-2027).
- Shilling et al. (2026) *The intrinsic luminosity–decay correlation in subsamples of GRB X-ray afterglows.* MNRAS, 552, stag1589
- Shilling et al. (2025) *Evidence for an intrinsic luminosity-decay correlation in GRB radio afterglows.* MNRAS, 542, 2421

Co-author

- Turnbull et al. (in prep.) *The Oates Relation: testing a structured jet origin.* In preparation (est. 2027).

  My contribution: The project is ongoing. So far, I have contributed to the method and approach, and provided insight in simulating a realistic population of light curves.
- Gupta et al. (2026) *Early Optical Follow-up of Gamma-Ray Bursts: The Critical Role of Robotic Telescopes.* Submitted to arXiv.
- Gupta et al. (2026) *Extremely Luminous Optical Afterglow of an Energetic Gamma-Ray Burst GRB 230204B.* ApJ, 997:246.

  My contribution: I contributed to the physical interpretation of this event in the context of radio afterglows.
- Miller et al. (2025) *The La Silla Schmidt Southern Survey.* PASP, 137:9.
- Levan et al. (2025) *Fast X-ray transient EP240315A from a Lyman-continuum-leaking galaxy at $z \approx 5$.* Nature Astronomy, 9:1375-1386.
- Gupta et al. (2024) *Deciphering the Physical Origin of GRB 240825A: A Long GRB Lacking a Bright Supernova.* Submitted to ApJ.